

Observing the Moon Lab

Due: XYZ

Learning Goals

1. Estimate a measurable physical quantity using a known reference scale, and justify the precision of that estimate.
2. Infer the motion of a distant body from observations.
3. Communicate observational results effectively and situate them within the historical record.
4. Interpret the absence of a detection as evidence rather than as failure.

Overview of the Lab

Since ancient times, numerous civilizations have marvelled at the Moon. In this lab, we ask you to do the same. We ask you to look at the moon every few days for a month and record some approximations of the position and shape of the moon. From this information, we then ask that you infer some information about the Moon.

Description of Expected Observations

Observe the moon at dusk (can be done alongside your "Sunset Lab") for one month at intervals of every 2-4 days. This should result in a *minimum* of 8 total observations, but ideally $\gtrsim 10$ observations total.

For each observation where you are able to see the Moon record in your lab notebook the following:

- Date and time of observation
- Measure the altitude of the moon. The altitude is the height, in degrees, a source resides above the horizon. The altitude can be estimated by the fact that the moon subtends approximately 0.5 a degree on the sky (so you should essentially record how many "moons" the moon is above the horizon and multiply by 2). Also approximate the uncertainty on this measurement of the altitude.
- Measure the azimuth of the moon. You can do this with a compass (on your phone, or a physical compass) by measuring the number of degrees East of North the Moon resides. If you do not have access to a compass, you can assume that the Sun has set at exactly West and use the fact that the moon subtends 0.5 a degree on the sky to estimate the azimuth. (It is not true that the Sun sets exactly in the West, but it means your measurement will only be off by some roughly constant systematic). Also approximate the uncertainty in your measurement of the azimuth.
- Sketch the moon and label it with the phase.

- Make notes of the observing conditions (is it cloudy? Is the weather bad? etc.).

Not being able to find the moon at dusk when you go looking for it is called a "non-detection". Non-detections are perfectly acceptable! Roughly half the nights you try to find the moon you will not see it in the sky at dusk (assuming perfect weather). For a non-detection of the moon record in your lab notebook the following:

- Date and time of observation
- Observing conditions (e.g., the weather)

We *will not* take points off if you do not have tools available (for example, if you use the size of the moon to estimate the altitude and azimuth) as long as your method of measurement is clearly stated in your lab notebook. Although, we do expect an appropriate estimation of the uncertainty on your measurement. For example, if you are using the size of the moon to estimate the altitude, you only know the true value within a few degrees.

Final Product (that you will submit)

1. Your lab notebook with all details that you have noted down (see above)
2. Three plots (each should have a short 2-3 sentence caption). These plots can be made in Excel or by hand.
 - (a) Altitude vs. Date. Represent non-detections as a downward arrow at the horizon.
 - (b) Azimuth vs. Date. Do not show non-detections.
 - (c) Altitude vs. Azimuth. Do not show non-detections.
3. A short report answering the below questions. The report should be "formal writing", written with paragraphs and complete sentences. Include citations whenever relevant (whichever format you prefer: MLA, APA, etc.). The write-up should be 1-2 pages, but this is not strict. Please just don't make us read too much!
 - (a) *Introduction/Background:* Begin with some background information about measurements of the moon. Historically, when are the first recorded observations of the moon? Who performed them and what did they record? How do they compare to the observations you made?
 - (b) *Methods:* How did you make your measurements? How did you estimate the uncertainty on your measurements? Explain and motivate your choices.
 - (c) *Interpretation:* What did you learn from your non-detections of the Moon? Why was it not observable in each of the cases of a non-detection?
 - (d) *Interpretation:* Look online for a calendar of moon phases. Using this calendar, which phases were you able to observe? Which phases were you not able to observe? Explain why.
 - (e) *Interpretation:* Infer the period of the orbit of the Moon around the Earth from your observations. Show all work and/or explain your method in complete sentences. Report your result in days. (Hint: You should use the altitude vs. azimuth plot that you created.)

Rubric

Topic	8-10 Points	5-7 Points	2-4 Points	0-1 Points
Observations	Observations were performed every 2-4 days (logged non-detections count as observations).	Observations were performed every 4-7 days.	Observations were performed bi-weekly.	Only a few, or less, observations were recorded.
Measurement & Uncertainty	Measurements and Uncertainties are appropriate and well justified.	Measurements are appropriate. Uncertainties are reported but are weakly motivated or are inappropriate for the measurement method used.	Measurements and/or uncertainties are reported inconsistently or without justification	Uncertainties and/or measurements are absent.
Lab Notebook	Notes are clear, detailed, well organized, and contain all of the information requested.	Notes contain all information requested but are disorganized.	Notes are disorganized and missing information.	Notes are non-existent or very limited.
Plots/Figures	All three figures are included, are interpretable/clear, and the results are logical.	All three figures are included but they are difficult to parse or the results are illogical.	Missing figures and/or the figures are impossible to read.	No figures are included.
Write-up (Quality)	The report is well written, easy to understand, includes appropriate citations	The write-up is poorly written, hard to understand, or missing citations	The report is poorly written, hard to understand, <i>and</i> missing citations	No report is included.
Write-up (Content)	All questions are answered thoroughly and correctly	All questions are answered but are missing details or are partially incorrect.	Some questions are not answered and/or a large majority of the questions are answered incorrectly.	Answers to most of the questions are missing and/or no report included.